



Generative-Temporal Modeling for Health Index Forecasting in Industrial Robotics: A GAN-Augmented Attention-TCN Approach under Data Scarcity Constraints

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Abstract: This article addresses short-horizon Health Index (HI) forecasting for industrial robots operating under severe data-scarcity constraints. In high-throughput photovoltaic manufacturing, preventive maintenance and PLC-bounded memory prevent the collection of complete run-to-failure trajectories, leaving critical degradation states under-represented in the training distribution. To mitigate this limitation, the proposed GAN-ATCN framework combines a conditional Generative Adversarial Network (cGAN) for targeted synthetic sequence generation with an Attention-based Temporal Convolutional Network (ATCN) for temporal health forecasting. The approach is validated on the NASA C-MAPSS benchmark and on an operational ST07 SCARA robot used in photovoltaic manufacturing. On the high-complexity FD004 subset, cGAN augmentation reduces the NASA Score by 58.73% compared with the non-augmented baseline. On the industrial robot case, the augmented model reduces HI forecasting RMSE by 47.7% despite limited PLC data and the absence of failure events. These results show that generative augmentation is most useful when operational complexity creates health-condition coverage gaps, supporting predictive maintenance and zero-defect manufacturing under realistic industrial constraints.

Keywords: Health Index Forecasting, Predictive Maintenance, Conditional GAN, Attention-TCN, Data Scarcity, Industrial Robotics.

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1. Introduction

Automated robotic systems in high-throughput manufacturing exhibit a structural paradox: the equipment most in need of predictive maintenance is precisely that for which sufficient degradation data is unavailable. In photovoltaic (PV) solar cell manufacturing, four-axis SCARA robots perform sub-millimetre pick-and-place operations at cycle rates below 1.2 seconds per cell. Progressive kinematic degradation reduces positional precision and propagates as defect cascades, including micro-cracks and production stoppages.

From a Lean manufacturing perspective, this degradation pattern directly generates three of the eight classical wastes identified by Ohno [23]: defects from positional deviation, over-processing from rework cycles, and waiting from unplanned stoppages. Within the Total Productive Maintenance (TPM) framework formalised by Nakajima [22], this quality-maintenance coupling constitutes a direct attack on Overall Equipment Effectiveness (OEE) through simultaneous Quality and Performance losses. Proactive maintenance that anticipates degradation before it propagates to the product level is therefore a cornerstone of both Lean and TPM philosophies and directly advances UN SDG 12 targets by reducing manufacturing waste and energy consumption [3, 20].

Deploying PdM within Prognostics and Health Management (PHM) is constrained by four structural barriers identified by Fink et al. [4]: data scarcity, distributional shifts across operating conditions, data confidentiality that prohibits cross-facility pooling, and the absence of failure labels under preventive maintenance. These barriers are compounded by hard infrastructure limits specific to the industrial context studied here. The ST07 SCARA robot operates under a PLC architecture that erases all positioning records on daily shutdown, bounding accessible historical data to a single operational session regardless of service duration. This PLC-bounded data retention constraint means the coverage gap in the joint health-condition space is not merely a question of sample size: critical degradation states are systematically absent from the training distribution by design, causing models to fail precisely where accurate prediction is most consequential [4, 34].

Despite recent advances in generative augmentation and temporal sequence modelling for PHM, two critical gaps remain. First, the conditions under which conditional GAN augmentation benefits continuous health state

regression, versus introducing distributional noise detrimental to generalisation, remain insufficiently characterised across benchmark conditions of varying operational complexity. Second, the combination of cGAN-based distribution engineering with an attention-based temporal architecture has received limited validation against the full set of industrial deployment constraints: PLC-bounded retention, absence of run-to-failure labels, and on-premise computational limits [4, 34].

To address these gaps, this study introduces the **GAN-ATCN framework**, integrating a conditional Generative Adversarial Network with an Attention-based Temporal Convolutional Network for short-horizon Health Index forecasting under industrial data scarcity.

Addressing this limited data trap requires tools capable of reconstructing missing distributional coverage without access to additional real observations. Conditional Generative Adversarial Networks [19] provide targeted synthesis of underrepresented degradation states by conditioning generation on specified health levels, directly targeting the coverage gaps imposed by PLC retention limits. Attention-based Temporal Convolutional architectures augmented with squeeze-and-excitation channel attention [12] and multi-head self-attention [28] provide the parallel processing efficiency required for real-time deployment within production cycle budgets. Together, these tools are not simply methodological choices: they are necessitated by the structural constraints of the industrial deployment context.

Classical RUL estimation requires failure timestamps as supervision signals, which are structurally absent under preventive maintenance. The Health Index (HI) formulation instead tracks continuous positioning performance relative to a nominal reference, providing a directly observable degradation signal without knowledge of the failure endpoint [13, 16]. This formulation is grounded in TPM’s autonomous maintenance pillar, which prescribes continuous condition monitoring as a first line of defence against degradation-induced quality losses [22]. A five-cycle prediction horizon aligns with the timescale at which degradation becomes quality-consequential, and SPC-aligned decision thresholds transform continuous HI forecasts into actionable maintenance triggers, extending Lean’s Jidoka principle—the authority to stop production before defects propagate upstream from product inspection to equipment condition [20, 24].

The overall framework proposed in this paper to address these challenges is illustrated in Figure 1.1, which provides a high-level

overview of the GAN-ATCN Health Index monitoring and forecasting system, encompassing data preprocessing, generative augmen-

tation, attention-based temporal forecasting, and decision-support integration.

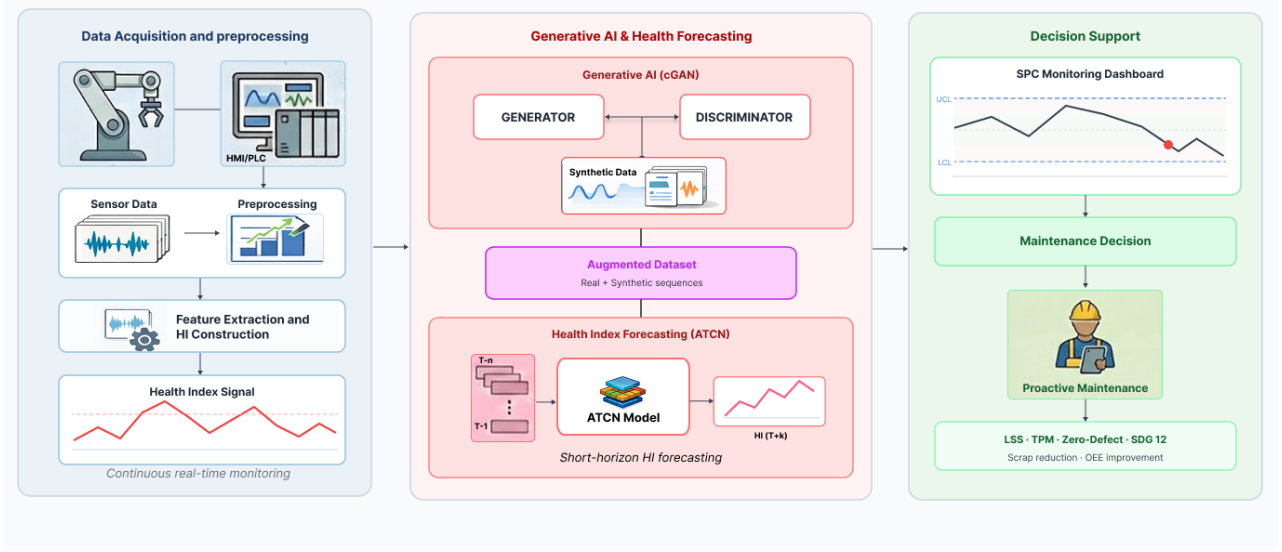


Figure 1.1: High-level overview of the GAN-ATCN Health Index monitoring and forecasting system.

Validation is conducted on the NASA C-MAPSS turbofan benchmark and on the operational ST07 SCARA robot under authentic PLC-bounded scarcity and zero-failure conditions. The research makes five primary contributions:

1. **Framing PdM under data scarcity** as a generative AI problem through targeted distribution engineering;
2. **Demonstrating consistent ATCN superiority** over LSTM baselines, with advantages scaling with operational complexity;
3. **Empirically characterising augmentation effectiveness** as a function of distributional coverage structure rather than data volume;
4. **Validating the approach on real industrial deployment**, achieving 47.7% RMSE reduction on the ST07 robot; and
5. **Introducing a short-horizon HI forecasting paradigm** that operationalises Lean manufacturing waste reduction and TPM's OEE pillars through SPC-based quality control for PV manufacturing.

The remainder of this article is organized accordingly.

2. Related Works

The GAN-ATCN framework sits at the intersection of four active research areas: deep learning for predictive maintenance (PdM)

and RUL estimation, temporal convolutional and attention-based sequence modelling, Generative AI augmentation for industrial prognostics, and PHM deployment under real industrial constraints. This section critically reviews each area, identifying the capabilities established by prior work and the limitations that collectively motivate the proposed approach.

2.1. From Classical Machine Learning to Deep Learning in Predictive Maintenance

Data-driven PdM has undergone a fundamental architectural shift over the past decade. Early approaches relied on shallow models, including Support Vector Machines, Random Forests, and k-Nearest Neighbor classifiers, applied to hand-engineered features derived from frequency-domain analysis or statistical signal moments. These methods demonstrated acceptable performance on low-dimensional, stationary fault classification tasks but encountered two compounding structural limitations as industrial datasets grew in complexity. First, the feature engineering burden was substantial: representations designed for one operating condition or fault mode transferred poorly to others. Second, shallow model capacity proved insufficient to capture the non-linear, multi-scale temporal dynamics of progressive equip-

ment degradation, with performance degrading systematically as operating condition heterogeneity increased [16, 32].

Deep learning addressed both limitations by learning hierarchical feature representations directly from raw sensor data, eliminating manual feature construction and enabling end-to-end optimisation from sensor input to maintenance decision [5]. The performance advantage over classical machine learning has been documented consistently and scales with operational complexity. Li et al. [17] established a deep convolutional neural network as a strong reference for RUL estimation on NASA C-MAPSS, achieving superior accuracy compared with existing approaches. Babu et al. [1] demonstrated that one-dimensional CNNs could extract informative local degradation features directly from raw sensor windows without frequency-domain preprocessing. Jin et al. [14] proposed a Time Series Transformer architecture for rotating machinery fault diagnosis, demonstrating superior fault-identification performance compared with CNN and RNN baselines. Across this body of work, a consistent finding emerges: the performance advantage of deep learning grows larger as the number of operating conditions, fault modes, and sensor channels increases [32], a scaling relationship that directly motivates the architecture choices made in this paper.

Despite these advances, a critical limitation persists: model performance is evaluated predominantly on complete benchmark datasets, and the impact of data scarcity on prognostic accuracy remains systematically underexplored. The implicit assumption that sufficient labelled training data is available does not hold in many real industrial deployments subject to data retention policies and preventive maintenance practices that yield no run-to-failure annotations. Bridging this gap requires generative augmentation strategies that the existing deep learning PdM literature has not yet addressed in a principled, deployment-validated way.

2.2. Temporal Convolutional Networks and Attention Mechanisms for Sequential Health Modelling

Among deep learning architectures for sequential sensor modelling, LSTM networks dominated the PdM literature for several years following their demonstrated capacity to model long-range temporal dependencies through gating mechanisms [11, 17]. How-

ever, LSTM architectures carry a structural limitation consequential for industrial deployment: sequential hidden-state propagation means inference latency scales linearly with sequence length, a bottleneck for monitoring systems that must produce health estimates within production cycle-time budgets under limited on-premises computational resources. Furthermore, while LSTM gating mitigates the vanishing gradient problem, it does not eliminate training instability for very deep stacks or long sequences under limited data, precisely the regime that characterises the data-constrained ST07 SCARA deployment.

TCNs, proposed by Bai et al. [2], offer a structurally superior alternative. By combining dilated causal convolutions with residual connections, TCNs achieve exponentially growing temporal receptive fields, fully parallel processing of timesteps, and stable gradient flow simultaneously. In a systematic comparison across eight sequence modelling benchmarks, Bai et al. [2] demonstrated that TCNs consistently match or outperform LSTMs and GRUs with greater computational efficiency and training stability. The Squeeze-and-Excitation (SE) module [12] and multi-head self-attention [28] address the remaining limitation of uniform feature weighting in standard TCNs: SE recalibrates channel-wise responses to selectively amplify degradation-relevant sensor dimensions, while self-attention dynamically focuses on the most informative timesteps for each prediction. Zhang et al. [31] integrated both mechanisms within a TCN backbone to form the ATCN architecture, demonstrating improvements over standalone TCN and LSTM baselines for C-MAPSS RUL estimation. The present work adapts this architecture for robot HI forecasting, introducing implementation modifications suited to industrial data scarcity, and evaluates it under Generative AI-augmented training.

2.3. Generative AI and GAN-Based Augmentation for Industrial Prognostics

Generative AI has emerged as a central paradigm for addressing data scarcity in industrial prognostics, with GAN-based models forming the dominant approach to date. The application of Generative AI to PHM has evolved through two distinct phases. The first phase addressed fault classification under class imbalance. Han et al. [6] applied Wasserstein GANs with gradient penalty to bearing fault classification, demonstrating that GAN-based minority-class augmentation

improves diagnostic accuracy under class imbalance. More recently, denoising diffusion probabilistic models have demonstrated superior distributional fidelity for time-series synthesis [10]; however, their iterative sampling process introduces inference latency incompatible with sub-second production-cycle constraints, making single-pass cGAN generation architecturally preferable for the edge-deployment context of this study. More recently, TimeGAN [30] introduced supervised autoregressive loss to preserve stepwise temporal dynamics in synthetic sequences; however, its requirement for complete degradation trajectories as supervision signals makes it unsuitable for the zero-failure PLC environment of this study, where such trajectories are structurally absent. These results established cGAN-based augmentation as a reliable strategy for categorical fault data; however, extending it to regression-based degradation forecasting, where synthetic samples must accurately reflect continuous health dynamics, requires additional capability that first-generation GANs were not designed to provide.

Conditional GANs (cGANs) [19] provide this capability by conditioning both generator and discriminator on continuous auxiliary variables, enabling targeted generation of degradation sequences at specified health levels. As surveyed by Wen et al. [29], data augmentation methods are particularly useful in time-series applications where labelled training data are limited, a regime directly analogous to the ST07 SCARA robot's PLC retention constraint of 5,000 cycles per day. Lu et al. [18] reported improvements from GAN-based sensor anomaly detection in robotic systems, providing a precedent for this approach in robot prognostics. Despite this growing body of evidence, the conditions under which cGAN-based augmentation is beneficial versus detrimental for continuous health state regression remain insufficiently characterised, and the combination of cGAN augmentation with an ATCN backbone has received limited validation under real manufacturing deployment constraints.

2.4. PHM Deployment Under Industrial Constraints and Quality Management Integration

Building on the barriers identified by Fink et al. [4], a persistent gap remains between algorithmic performance on controlled benchmarks and the requirements of real industrial

deployment, including limited access to historical data, on-premises computing requirements, and multi-condition operational variability. The ST07 SCARA case study presented in Section 5 is explicitly evaluated against all four criteria.

The HI forecasting formulation is grounded in the well-established PHM literature for systems in which failure labels are structurally unavailable. Jardine et al. [13] formalised the condition-based maintenance framework, thereby motivating continuous health-state estimation. Lei et al. [16] established that deviation-based proxies provide valid degradation signals for mechanical systems with measurable kinematic outputs. The integration of PHM with quality management has emerged as a strategically important research direction: Zhong et al. [33] and Kusiak [15] argued that the traditional separation between maintenance and quality engineering is untenable in highly automated environments, while Psarommatidis et al. [24] formalised this connection in a zero-defect manufacturing framework that integrates predictive maintenance triggers directly into the quality control decision loop.

Taken together, this body of work identifies four open issues that motivate the approach proposed in Section 3. First, the conditions under which cGAN-based augmentation is beneficial for continuous HI and RUL regression remain insufficiently characterised. Second, the ATCN architecture has received limited evaluation under Generative AI-augmented training or in the context of authentic industrial data scarcity. Third, the interplay between multi-condition variability and augmentation effectiveness requires controlled investigation. Fourth, the cGAN-ATCN combination has received limited validation against the full set of industrial deployment barriers identified by Fink et al. [4] and Zonta et al. [34]. The GAN-ATCN framework is designed to address all four issues within a single, deployment-validated study.

3. Methodology

The GAN-ATCN framework comprises three components: (1) conditional GAN-based synthetic data generation, (2) an Attention-based Temporal Convolutional Network (ATCN) for health forecasting, and (3) a hybrid training protocol combining real and synthetic data. Two formulations are addressed: (i) RUL prediction for benchmark evaluation on NASA C-MAPSS, serving as a controlled validation proxy, and (ii) short-horizon Health Index (HI)

forecasting for industrial deployment on the ST07 SCARA robot, which constitutes the pri-

mary operational objective. The workflow is illustrated in Figure 3.1.

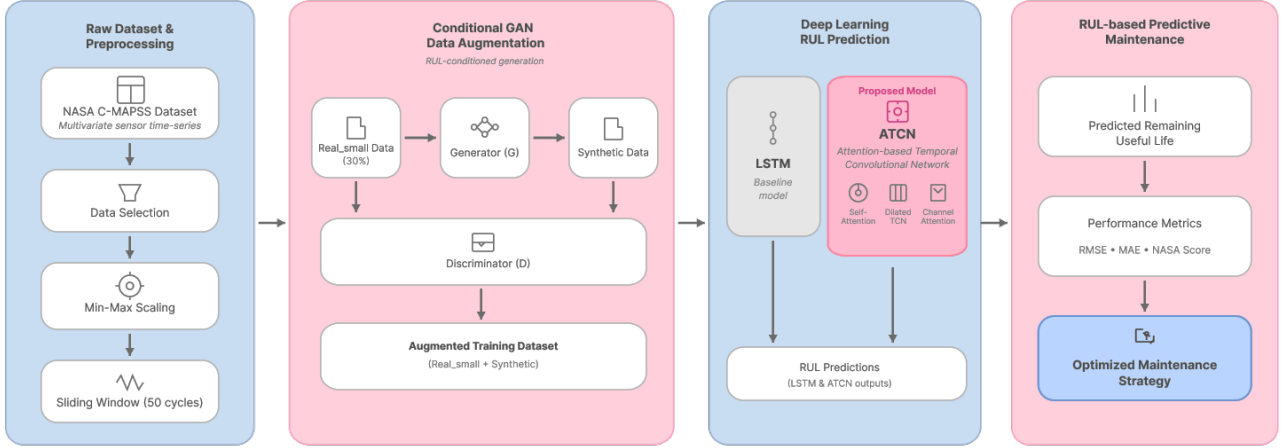


Figure 3.1: GAN-ATCN workflow: preprocessing and sequence construction, cGAN-based synthetic data generation, and ATCN-based short-horizon HI forecasting

3.1. Problem Formulation

The RUL prediction task is formulated as a time-series regression problem. Let

$$\mathbf{X}_{1:t} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t\}, \quad \mathbf{x}_t \in \mathbb{R}^N$$

represent a multivariate sequence of sensor observations, where N is the number of sensor channels. A sliding window of length W is defined as

$$\mathbf{X}_t = [\mathbf{x}_{t-W+1}, \dots, \mathbf{x}_t] \in \mathbb{R}^{W \times N}$$

The objective is to learn a non-linear mapping function $f : \mathbb{R}^{W \times N} \rightarrow \mathbb{R}$ that predicts the health state $\hat{y}_t$, corresponding either to RUL or HI:

$$\hat{y}_t = f(\mathbf{X}_t; \theta)$$

where θ denotes the model parameters. The optimal parameters are obtained by minimising the mean squared error over M training samples:

$$\theta^* = \arg \min_{\theta} \frac{1}{M} \sum_{i=1}^M (y_i - \hat{y}_i)^2$$

For C-MAPSS benchmark evaluation (Sections 4.1-4.4), the ground truth y_t follows a piecewise linear degradation model:

$$y_t = \begin{cases} R_{early}, & t < t_{deg}, \\ R_{early} - (t - t_{deg}), & t \geq t_{deg} \end{cases}$$

where $R_{early} = 125$ cycles is the maximum RUL cap, and t_{deg} is the degradation onset point. This formulation prevents the network from learning spurious early-life gradients when sensor readings are physically indistinguishable from nominal operation. For the ST07 industrial deployment (Section 5),

the target is the future Health Index at a 5-cycle prediction horizon, a continuous degradation metric on $[0, 1]$ where values near unity indicate healthy operation. In the face of data scarcity, the cGAN component addresses coverage gaps by targeting synthetic augmentation of underrepresented health-condition regions prior to ATCN training.

3.2. Conditional GAN Architecture

The cGAN [19] generates synthetic multivariate sensor sequences conditioned on target health levels. Generator G and discriminator D are trained adversarially:

$$V(D, G) = \mathbb{E}_{(\mathbf{x}, c)} [\log D(\mathbf{x}, c)] + \mathbb{E}_{(\mathbf{z}, c)} [\log (1 - D(G(\mathbf{z}, c), c))]$$

$$(G^*, D^*) = \arg \min_G \arg \max_D V(D, G)$$

The generator $G(\mathbf{z}, c)$ maps latent noise $\mathbf{z} \in \mathbb{R}^{32}$ and health condition c to synthetic sequences $\tilde{\mathbf{x}} \in \mathbb{R}^{50 \times 14}$ via two Dense layers (128 units, ReLU), reshaped to (seq len, 64), followed by two 1D convolutional layers (64 and 32 filters, kernel size 3, ReLU) and a tanh-activated output convolution, ensuring outputs lie in $[-1, 1]$. The discriminator $D(\mathbf{x}, c)$ processes sequence-condition pairs through two 1D convolutional layers (64 filters, kernel size 3), global average pooling, and a sigmoid output. Training runs for 40 epochs using the Adam optimizer (*learning rate 0.0001, default Adam β parameters*).

3.3. Attention-Based Temporal Convolutional Network (ATCN)

The ATCN architecture proposed in this work is inspired by Zhang et al. [31]. It integrates an Improved Self-Attention mechanism, dilated causal TCN blocks, and Squeeze-and-Excitation recalibration within a unified temporal modelling pipeline. The present implementation introduces a dense input projection layer. It applies SE recalibration globally following the TCN stack, adapting the original design to the data scarcity and industrial deployment constraints of this study.

3.3.1 Input Projection and Self-Attention

Prior to temporal feature extraction, the input sequence is projected to the model dimension (64 units) via a dense layer. An Improved Self-Attention (ISA) block then applies multi-head self-attention (2 heads) [28] with layer normalisation and a position-wise feed-forward network (128 units, ReLU), providing global context before local temporal feature extraction.

3.3.2 Temporal Feature Extraction

Four TCN blocks [2] with dilation rates $d \in \{1, 2, 4, 8\}$ and kernel size $k = 3$ form the temporal backbone. Each block applies two dilated causal convolutions with 64 filters and ReLU activation, with residual connections for gradient stability [7]. The receptive field spans 61 timesteps, fully covering the 50-cycle input window:

$$\begin{aligned} RF &= 1 + \sum_{\ell=1}^4 2d_{\ell}(k-1) \\ &= 1 + 2(1 + 2 + 4 + 8)(2) \\ &= 61 \end{aligned}$$

A Squeeze-and-Excitation (SE) module [12] is applied globally following the TCN stack, recalibrating channel-wise feature responses via global average pooling and a sigmoid-gated bottleneck to selectively amplify degradation-relevant sensor dimensions.

3.3.3 Output Layer and Training

Global average pooling over the attention-weighted sequence feeds a dense layer (64 units, ReLU) and a linear output unit.

The ATCN is trained using the Adam optimiser with a learning rate of 0.001 and a batch size of 256, with early stopping (patience 8) applied to the validation Huber loss ($\delta = 5.0$) to

prevent overfitting under data-scarce conditions. Hyperparameters were set following established TCN conventions [2], with the architecture depth and dilation rates selected to ensure sufficient receptive-field coverage of the 50-cycle input window. All hyperparameters were held constant across experimental conditions to isolate the effect of augmentation. A ReduceLROnPlateau scheduler halves the learning rate when validation loss plateaus for 4 consecutive epochs (minimum lr = 1e-6)

3.4. Hybrid Training Protocol

Sensors with near-zero variance (standard deviation below 0.01) are removed, and the remaining channels are normalised to $[-1, 1]$ using min-max scaling fitted exclusively on the training data. For C-MAPSS, RUL labels are clipped at 125 cycles [8]. Temporal sequences are constructed via 50-cycle sliding windows with unit stride.

C-MAPSS data scarcity is simulated by retaining 30% of training units via unit-level sampling, reducing the diversity of degradation trajectories available during training. The 30% retention rate was selected to induce meaningful coverage gaps in the training distribution while preserving model training stability, enabling controlled evaluation of augmentation effectiveness under simulated scarcity conditions.

After preprocessing, the cGAN is trained for 40 epochs on real training sequences. The generator then produces synthetic sequences of the same size as the real training set, with conditioning labels randomly resampled from the training RUL distribution. Two ATCN configurations are compared: a baseline using only real sequences, and an augmented model using real-plus-synthetic data, with all other variables held constant. Evaluation is conducted exclusively on real test sets. Each condition is evaluated on the real test set with a fixed random seed (42) for reproducibility.

4. Experimental Validation

This section validates the GAN-ATCN framework on the NASA C-MAPSS turbofan benchmark across two sequential stages: establishing ATCN performance relative to an LSTM baseline, and evaluating the effectiveness of cGAN augmentation under simulated data scarcity.

4.1. NASA C-MAPSS Dataset

The C-MAPSS dataset [25] provides multivariate sensor sequences from simulated turbofan engines undergoing progressive degradation. Its four subsets vary systematically in operational complexity, enabling controlled evaluation of how complexity moderates augmentation effectiveness, the primary experimental

question of Section 4.4. Table 4.1 summarises the subsets; Figure 4.1 illustrates the turbofan architecture.

All datasets undergo identical preprocessing per Section 3.4: variance filtering, retaining 14 active channels, RUL clipping at 125 cycles, min-max normalisation, and 50-cycle sliding windows with unit stride. Data scarcity is simulated by retaining 30% of training units.

Table 4.1: C-MAPSS Dataset Characteristics

Dataset	Operating Conditions	Fault Modes	Training Units	Test Units	Complexity
FD001	1 (Sea level)	1 (HPC)	100	100	Low
FD002	6 (Altitude $\times$ Mach $\times$ TRA)	1 (HPC)	260	259	Medium
FD003	1 (Sea level)	2 (HPC + Fan)	100	100	Medium
FD004	6 (Altitude $\times$ Mach $\times$ TRA)	2 (HPC + Fan)	248	249	High

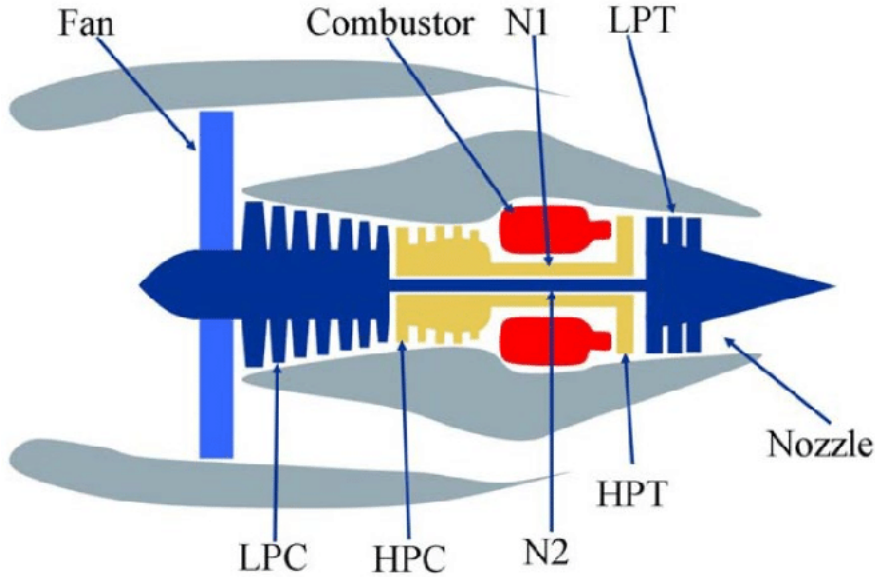


Figure 4.1: Turbofan Engine Architecture and Sensor Measurement Locations

4.2. Performance Metrics

Four complementary metrics are employed to evaluate model performance across both the NASA C-MAPSS benchmark and the ST07 industrial deployment. Each metric captures a distinct aspect of predictive accuracy and operational reliability.

Root Mean Square Error (RMSE) quantifies average prediction accuracy:

$$RMSE = \sqrt{\frac{1}{N_s} \sum_{i=1}^{N_s} (y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE) provides a linear, scale-interpretable measure of average error that is less sensitive to outliers than RMSE:

$$MAE = \frac{1}{N_s} \sum_{i=1}^{N_s} |y_i - \hat{y}_i|$$

Coefficient of Determination (R^2) measures the proportion of variance in the true values explained by the model. A value of 1 indicates a perfect fit, while 0 indicates no explanatory power:

$$R^2 = 1 - \frac{\sum_{i=1}^{N_s} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N_s} (y_i - \bar{y})^2}$$

NASA Asymmetric Scoring Function [25] penalises late predictions, where the model overestimates remaining useful life, more severely than early predictions. This asymmetry reflects the greater operational risk of unexpected failures. The prediction error is defined as

$$d_i = \hat{y}_i - y_i$$

and the asymmetric penalty is

$$s(d_i) = \begin{cases} \exp\left(-\frac{d_i}{13}\right) - 1, & d_i < 0, \\ \exp\left(\frac{d_i}{10}\right) - 1, & d_i \geq 0 \end{cases}$$

Negative errors correspond to early predictions, while non-negative errors correspond to late predictions. The total NASA score is then

$$\text{NASA Score} = \sum_{i=1}^{N_s} s(d_i)$$

Where N_s is the number of evaluated observations, y_i is the actual value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the actual values, and d_i is the prediction error at observation i .

A lower NASA Score indicates fewer and less severe late predictions. All four metrics are reported jointly throughout Section 5, as they capture complementary aspects of model behaviour: RMSE and MAE quantify the average error magnitude, R^2 reflects overall goodness of fit, and the NASA Score isolates operationally critical late-prediction failures that low average-error metrics may otherwise obscure. This dual-reporting approach ensures

that a model with low average error but occasional catastrophic overestimates of remaining useful life is correctly identified as operationally unacceptable.

4.3. Baseline Comparison: LSTM vs. ATCN

The LSTM baseline comprises three bidirectional layers with 64 units per direction and an identical prediction head, matching the ATCN in total parameter count. Both models use identical optimisation settings (Adam, learning rate 0.001, batch size 256, early stopping with patience 8). The LSTM trains on full real datasets without scarcity simulation to establish upper-bound performance. Table 4.2 reports the comparison.

Table 4.2: LSTM vs. ATCN Performance Across C-MAPSS Subsets

Dataset	Model	RMSE (mean $\pm$ std)	NASA Score (mean $\pm$ std)	Δ RMSE	Δ NASA
FD001	LSTM	17.16 $\pm$ 6.64	970.14 $\pm$ 1,991.73	—	—
FD001	ATCN	15.82 $\pm$ 0.69	410.84 $\pm$ 100.52	−7.8%	−57.7%
FD002	LSTM	39.96 $\pm$ 13.15	63,818.53 $\pm$ 58,033.43	—	—
FD002	ATCN	32.35 $\pm$ 1.35	36,755.25 $\pm$ 20,883.20	−19.0%	−42.4%
FD003	LSTM	16.27 $\pm$ 0.57	952.40 $\pm$ 416.22	—	—
FD003	ATCN	15.92 $\pm$ 0.40	551.35 $\pm$ 173.03	−2.1%	−42.1%
FD004	LSTM	42.97 $\pm$ 10.94	305,292.26 $\pm$ 419,876.08	—	—
FD004	ATCN	33.34 $\pm$ 2.27	20,428.34 $\pm$ 10,020.49	−22.4%	−93.3%

The ATCN achieves consistent improvements over the LSTM across all four subsets, with RMSE reductions scaling from 2.1% on the single-condition FD003 to 22.4% on the most complex FD004. Training stability improves markedly: run-to-run RMSE standard deviation is four to eight times lower than the LSTM (e.g., FD001: 0.69 vs. 6.64), attributable to TCN parallel processing and residual connections. The 93.3% reduction in NASA score on FD004, a 14.9-fold decrease in the operational risk penalty, indicates that the ATCN specifically reduces the catastrophic late predictions that are most severely penalised under multi-condition degradation. The performance advantage consistently scales with operational complexity, confirming the ATCN as the ap-

propriate backbone for subsequent augmentation experiments.

4.4. Effect of GAN-Based Data Augmentation

Under the 30% scarcity condition, a baseline ATCN trained on the reduced real dataset (real_small) is compared against an augmented model trained on combined real-plus-synthetic sequences. This retention rate was selected as a controlled experimental setting to simulate data scarcity by reducing the diversity of available training trajectories. All other variables are held constant. Table 4.3 reports the comparison.

Table 4.3: Effect of Generative AI-Based Data Augmentation on ATCN Performance

Dataset	Training Data	RMSE	Δ RMSE vs. real_small	NASA Score	Δ NASA vs. real_small
FD001	real_small	15.77	—	346.90	—
FD001	real+synthetic	15.83	+0.42%	362.54	+4.51%
FD002	real_small	36.56	—	127,361.77	—
FD002	real+synthetic	35.91	-1.77%	74,933.49	-41.16%
FD003	real_small	18.06	—	689.45	—
FD003	real+synthetic	19.54	+8.22%	896.79	+30.07%
FD004	real_small	39.72	—	167,306.50	—
FD004	real+synthetic	39.89	+0.43%	69,042.00	-58.73%

Augmentation benefits are strongly conditional on operational complexity. For FD002 and FD004, both with six operating conditions, augmentation reduces the NASA score by 41.16% and 58.73%, respectively, while leaving RMSE within $\pm 2\%$, a range within measurement uncertainty under fixed-seed evaluation. This pattern reflects an operationally valuable tradeoff: augmentation shifts predictions toward more conservative estimates, reducing dangerous late predictions without substantially degrading average accuracy. For FD001 and FD003 single-condition subsets, augmentation provides negligible benefit or actively degrades performance, with the most severe degradation on FD003 (RMSE +8.22%, NASA score +30.07%). This pattern is interpretable through distributional regularisation. Under single-condition operation with 30% training units, real data already adequately covers the degradation space, and additional stochastic variability introduced by synthetic injection increases noise without adding coverage. Under six-condition operation, 30% unit retention sys-

tematically underrepresents each operating regime; cGAN augmentation improves robustness to unseen health-condition combinations at test time, yielding the observed NASA Score reductions despite limited temporal fidelity in individual synthetic sequences.

4.5. Qualitative Analysis: Prediction Stability

Figures 4.2 and 4.3 provide a qualitative illustration for FD004. The augmented model produces smoother RUL trajectories with substantially fewer large deviations during late degradation (Figure 4.2). Residual analysis (Figure 4.3) confirms that the baseline exhibits concentrated, large positive residuals at low RUL, the errors most severely penalised by the NASA scoring function. At the same time, the augmented model shows markedly reduced residual scatter and fewer extreme late predictions in the critical end-of-life region, consistent with the 58.73% improvement in the NASA score reported in Table 4.3.

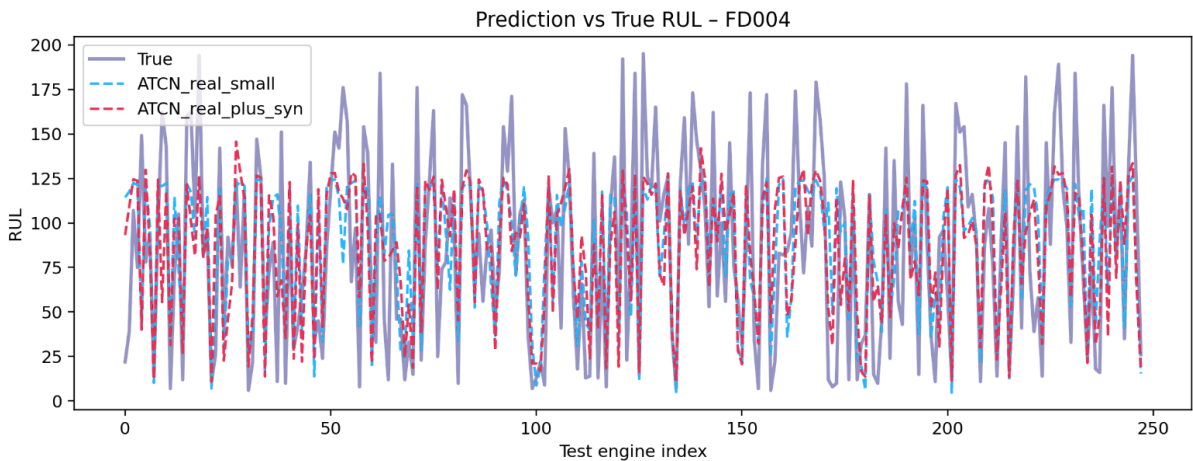


Figure 4.2: Predicted vs. True RUL (FD004)

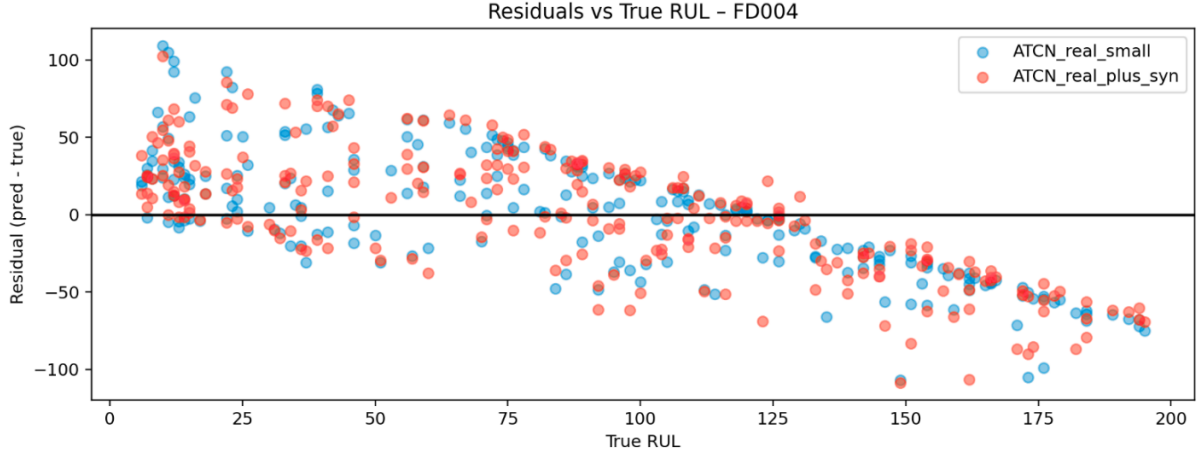


Figure 4.3: Residuals vs. True RUL (FD004)

5. Industrial Case Study

This section validates the GAN-ATCN framework on an operational ST07 four-axis SCARA robot deployed in photovoltaic solar panel manufacturing. The validation addresses three critical gaps from the C-MAPSS experiments: deployment under authentic data scarcity (short observation windows, zero failure events), health index forecasting for high-reliability equipment under preventive maintenance, and feasibility under industrial computing constraints.

5.1. Operational Context

The ST07 SCARA robot examined in this study operates within a photovoltaic module manufacturing facility whose core mission is high-throughput, defect-free cell assembly across multiple automated production stages. The facility required a degradation-forecasting solution capable of anticipating positioning failures before they propagate into product-level quality defects, deployable within existing on-premises infrastructure without additional hardware investment or cloud dependency. The GAN-ATCN framework was developed to meet these requirements directly, providing short-horizon health index forecasting under authentic data scarcity conditions and within the computational constraints of the production environment.

The ST07 operates on a tabber-stringer production line, performing high-frequency pick-and-place transfers of fragile silicon solar cells from a computer-vision inspection station to an automated conveyor for downstream soldering. Cycle times range from 0.8 to 1.2 seconds per cell. Sub-millimeter positional tolerances across all four degrees of freedom (axes

X, Y, Z, R) are mandatory: deviations beyond specification increase the probability of micro-cracks from mechanical stress and macro-cracks requiring full panel rework, detectable at downstream electroluminescence. At production scales of thousands of cells per shift, sustained positioning degradation directly reduces yield and overall equipment effectiveness (OEE). Figure 5.1 shows the industrial robot system used in this case study.

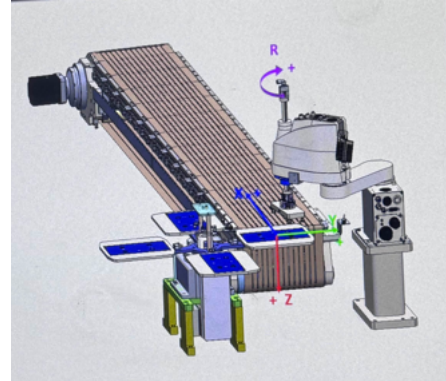


Figure 5.1: ST07 SCARA Industrial Robot System

Data processing occurs on-premises; proprietary security requirements prohibit cloud transmission, and the system operates within limited on-premises computational resources. Health forecasts must be completed within production cycle time budgets for synchronous monitoring. The facility follows a preventive maintenance policy that includes scheduled recalibration, lubrication, cleaning, and component servicing, resulting in zero catastrophic failure events during normal production. This precludes the accumulation of run-to-failure trajectories required by supervised RUL estimation.

Data acquisition was conducted over a 2.5-hour continuous monitoring window, capturing 5,000 operational cycles. The PLC retains

positioning records for up to 5,000 cycles per day and erases all records during daily shut-down, preventing longitudinal analysis across production days. This retention architecture imposes a hard bound on the amount of accessible historical data, regardless of the robot's service duration.

5.2. Health Index Construction

Health index forecasting is adopted because it tracks robot positioning performance without requiring failure endpoint labels [16, 26]. Nine raw features are extracted per cycle from the PLC: eight positioning deviation signals (DifPickX/Y/Z/R, DifPlaceX/Y/Z/R) across four kinematic axes, and CycleTime (Table 5.1). A binary operating-condition flag derived via median-based CycleTime partitioning (Section 5.3) is appended, yielding 10 input channels.

Table 5.1: ST07 SCARA Robot Positioning Deviation Features

Feature	Description	Unit
DifPickX	Pick position error (X-axis)	mm
DifPickY	Pick position error (Y-axis)	mm
DifPickZ	Pick position error (Z-axis)	mm
DifPickR	Pick rotation error	degrees
DifPlaceX	Place position error (X-axis)	mm
DifPlaceY	Place position error (Y-axis)	mm
DifPlaceZ	Place position error (Z-axis)	mm
DifPlaceR	Place rotation error	degrees
CycleTime	Operation execution time	seconds

The HI construction proceeds through four transformations, grounded in documented robot failure mechanisms, cumulative drive wear, mechanical backlash, bearing degradation, and encoder drift, all of which manifest as increased positioning errors [9, 27]. First, the eight deviation signals are aggregated via the Euclidean norm:

$$DevMag_t = \|d_t\|_2 = \sqrt{\sum_{j=1}^8 (d_{j,t})^2}$$

Second, the magnitude is normalised to $[0, 1]$ using min-max scaling fitted on training data:

$$DevNorm_t = \frac{DevMag_t - \min(DevMag)}{\max(DevMag) - \min(DevMag)}$$

Third, inversion maps low positioning error to high health:

$$HI_t^{raw} = 1 - DevNorm_t$$

Fourth, a 20-cycle rolling mean suppresses high-frequency noise while preserving genuine degradation trends:

$$HI_t = \frac{1}{20} \sum_{i=t-19}^t HI_i^{raw}$$

The resulting HI has a mean of 0.642, a standard deviation of 0.112, and a range of $[0.349, 1.000]$ over the 5,000-cycle window (Figure 5.2). The signal exhibits non-monotonic dynamics: rather than a steady decline, it alternates between nominal performance and recurrent low-health episodes representing elevated positioning error. These episodes are precisely the states that short-horizon forecasting seeks to anticipate, providing operators with sufficient lead time to schedule preventive action before positioning degradation propagates to product-level quality defects.

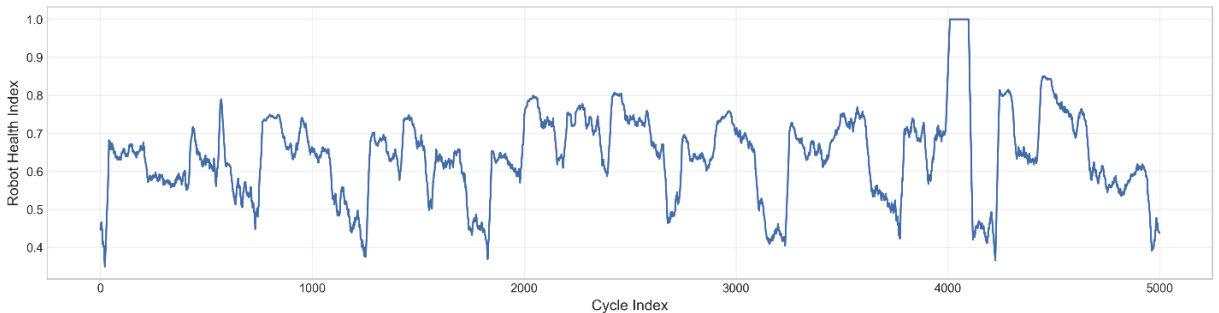


Figure 5.2: Temporal Evolution of the Robot Health Index (RHI) over 5,000 operational cycles

For operational decision support, the HI maps to a maintenance decision framework consistent with Statistical Process Control (SPC) principles [21]: $HI > 0.8$ indicates nominal operation; $0.5 \leq HI \leq 0.8$ warrants scheduling at the next planned maintenance window; $HI < 0.5$ requires priority inspection. This framework extends condition monitoring upstream from product quality outcomes to equipment state, aligning with Zero-Defect Manufacturing objectives [24].

5.3. Operating Condition Detection

Median-based partitioning of CycleTime identifies two natural operating regimes: a slow-cycle regime (Condition 0, mean 1.37 s, 50.2% of observations) and a fast-cycle regime (Condition 1, mean 0.74 s, 49.8%). Cycle time variation arises from downstream synchronisation dynamics rather than deliberate modification of robot control parameters; both regimes demand identical sub-millimeter positioning precision. A binary condition flag is appended to the feature set. The near-balanced 50-50 split mitigates class imbalance as a confounding factor and provides a structural analogue to the multi-condition FD002 and FD004 subsets evaluated in Section 4.

5.4. GAN-ATCN Implementation

The same architectural configuration described in Section 3 is applied without modifi-

cation. Input dimensionality is adjusted from 14 channels (C-MAPSS) to 10 channels (ST07) to reflect dataset-specific features. Data is segmented into 50-cycle sliding windows with unit stride, yielding 4,945 overlapping sequences. Each window is labelled with the HI at a 5-cycle prediction horizon, corresponding to approximately 5–7 seconds at nominal cycle rates, sufficient lead time for operators to respond before degraded positioning accumulates quality risk.

The dataset is partitioned chronologically: 80% (cycles 1–4,000) for training, 20% (cycles 4,001–5,000) for testing, preserving causal structure. The cGAN generates synthetic sequences conditioned on HI values resampled randomly from the training HI distribution. All evaluations are conducted on the real test set.

5.5. Health Forecasting Performance

Table 5.2 presents quantitative forecasting performance on four metrics: RMSE (primary accuracy indicator), MAE, R^2 , and NASA asymmetric score. Due to the bounded HI domain $[0,1]$, the NASA scoring function is interpreted only as a relative indicator of underestimation bias. Rolling-average smoothing during HI construction introduces temporal autocorrelation that may inflate R^2 ; RMSE is therefore the primary indicator throughout.

Table 5.2: ST07 SCARA Robot Health Forecasting Performance (5-cycle prediction horizon)

Configuration	RMSE	MAE	R^2	NASA	Δ RMSE	Δ NASA
ATCN (real only)	0.0368	0.0141	0.8895	6.3	Baseline	Baseline
ATCN (real+GAN)	0.0192	0.0148	0.9698	5.8	-47.7%	-7.6%

The baseline ATCN achieves RMSE of 0.0368 and R^2 of 0.8895, confirming that 50-cycle windows capture sufficient health dynamics for meaningful short-horizon prediction. GAN augmentation reduces RMSE from 0.0368 to 0.0192, a 47.7% reduction that nearly halves average prediction error, and increases R^2 from 0.8895 to 0.9698. The NASA score improves directionally from 6.3 to 5.8 (7.6%). The smaller proportional gain in the NASA score relative to RMSE reflects that GAN augmentation principally improves representation of moderate health states underrepresented in the single training window, producing broad accuracy gains captured by RMSE,

while the rarest health transitions are only partially resolved by synthetic coverage expansion.

Qualitatively, the baseline model captures the global HI trend but exhibits systematic lag during rapid transitions characterising low-health episodes, precisely the periods of highest quality risk (Figure 5.3). The augmented model substantially reduces this lag, enabling earlier and more reliable maintenance alerts. Figure 5.4 reveals systematic overestimation bias in the baseline at high HI values, attributable to the scarcity of near-pristine training examples. cGAN augmentation with random resampling from the training HI distribu-

tion substantially reduces this bias, producing tighter scatter around the ideal diagonal with reduced dispersion at both health extremes.

Table 5.3 reports the same comparison stratified by operating condition.

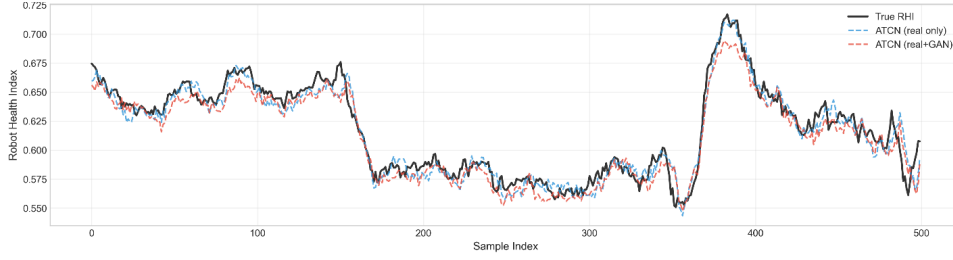


Figure 5.3: Health index forecasting – time series comparison

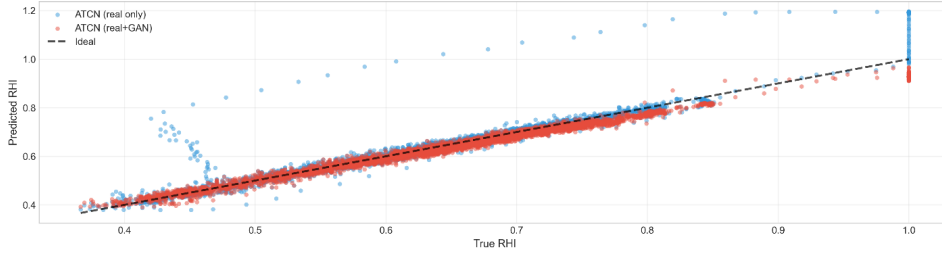


Figure 5.4: Prediction Accuracy, Predicted vs. True HI

Table 5.3: Performance Stratified by Operating Condition

Condition	Configuration	RMSE	Improvement
0 (Slow-cycle)	ATCN (real only)	0.0391	Baseline
0 (Slow-cycle)	ATCN (real+GAN)	0.0202	-48.3%
1 (Fast-cycle)	ATCN (real only)	0.0345	Baseline
1 (Fast-cycle)	ATCN (real+GAN)	0.0182	-47.2%

Performance improvements are consistent across both operating regimes, with RMSE reductions of approximately 47–48%, indicating uniform generalization improvement. This result aligns with the multi-condition augmentation mechanism identified in Section 4, confirming that cGAN-based coverage expansion generalises from controlled benchmark conditions to the industrial deployment setting.

6. Discussion

The results presented in Sections 4 and 5 reveal a consistent and interpretable mechanism: cGAN-based augmentation improves forecasting accuracy when the training distribution structurally underrepresents the joint health-condition space, and is neutral or detrimental when real data already provides adequate coverage. These findings suggest that augmentation serves as a distributional-

correcting mechanism, rather than a universally beneficial data-expansion strategy.

6.1. Mechanistic Basis of Augmentation Effectiveness

The structure of the training distribution governs the effectiveness of augmentation. In single-condition settings (FD001, FD003), 30% unit retention is sufficient to preserve coverage of degradation trajectories, as all units share a common operating regime. Under these conditions, synthetic injection introduces off-manifold variability that disrupts the existing distributional structure, yielding detrimental noise rather than informative coverage expansion. In contrast, multi-condition settings (FD002, FD004) exhibit systematic underrepresentation across operating regimes under the same retention level, exposing the model to unseen health-condition combinations at test time. Conditional gener-

ation with random resampling from the training distribution provides additional sequence diversity, improving generalisation and reducing high-risk prediction errors.

The ST07 deployment provides convergent evidence under a different form of scarcity: temporal underrepresentation within a single asset. The PLC retention limit constrains the training window to 4,000 cycles dominated by high-health states, with degradation transitions sparsely observed. Synthetic sequences conditioned via random resampling from the training HI distribution introduce additional sequence diversity, improving forecast fidelity at both health extremes and reducing prediction lag during rapid transitions. The resulting 47.7% RMSE reduction, substantially larger than gains observed on C-MAPSS, reflects the greater severity of temporal scarcity. Overall, augmentation effectiveness scales with distributional coverage gaps, not raw data volume.

6.2. Architectural Suitability for Industrial Deployment

The ATCN's consistent superiority over the LSTM baseline, with performance advantages increasing under higher operational complexity, reflects properties well-aligned with industrial constraints. Unlike LSTM architectures, whose hidden-state updates require sequential evaluation of h_t that depends on h_{t-1} , the ATCN's dilated causal convolutions decouple temporal processing from execution order, enabling fully parallel inference within the 0.8–1.2-second production cycle time without reliance on external computing resources. In addition, substantially improved training stability, with run-to-run variance four to eight times lower than that of an LSTM, is critical under PLC-bounded data conditions, where unstable models would undermine deployment reliability.

The attention mechanisms provide complementary benefits. SE channel recalibration enhances sensitivity to degradation-relevant sensor dimensions, while multi-head self-attention enables adaptive focus on short temporal segments that drive near-term degradation. Together, these properties support robust short-horizon forecasting, where accurate anticipation of rapid transitions is more critical than long-term trend estimation.

6.3. Practical Implications for Industry

The GAN-ATCN framework operates as a predictive decision-support system rather than a monitoring tool. By forecasting near-future health states over a 5-cycle horizon (approximately 5–7 seconds), it enables intervention before positioning degradation propagates into product-level defects.

From an industrial engineering perspective, this anticipatory capability directly impacts key performance metrics. Early detection of degradation reduces scrap and rework associated with micro-crack and macro-crack defects, lowering the Cost of Poor Quality (COPQ) and supporting Lean waste reduction. The integration of SPC-aligned thresholds (HI > 0.8 nominal; 0.5–0.8 moderate; < 0.5 priority) transforms continuous forecasts into actionable control decisions, effectively extending Statistical Process Control upstream from product inspection to equipment condition.

These improvements translate into measurable gains in Overall Equipment Effectiveness (OEE). Within a 5–7 second predictive horizon, the framework is specifically optimised to intercept short-fuse degradation pathways: it mitigates Quality losses by halting feed mechanisms before positioning deviation generates micro-crack defects, and reduces Performance losses from jam-induced minor stoppages. Impact on long-term Availability scheduling falls outside the scope of this prediction horizon. The 47.7% RMSE reduction improves the reliability of threshold-based decisions, reducing both false alarms and missed detections.

The on-premise deployment architecture further supports industrial applicability by operating within data confidentiality and infrastructure constraints, without reliance on cross-facility data sharing or cloud-based computation.

7. Conclusion

This paper presented the GAN-ATCN framework, combining conditional generative augmentation with an attention-based temporal convolutional network for short-horizon Health Index forecasting under industrial data scarcity. Generative augmentation proves beneficial under multi-condition scarcity, achieving a 58.73% reduction in NASA Score on FD004 and a 47.7% reduction in RMSE on the ST07 SCARA robot, while remaining neutral or detrimental when distributional cover-

age is already sufficient. The framework integrates SPC-aligned decision thresholds with continuous HI forecasting, supporting Zero-Defect Manufacturing and OEE improvement within Lean manufacturing constraints. Validation is conducted on a single asset within a single manufacturing process, limiting generalisation claims. The cGAN assumes distributional stationarity, making it susceptible to long-term concept drift caused by component replacement or process changes. The framework produces deterministic point estimates that fail to decompose predictive variance into aleatoric uncertainty arising from mechanical noise and epistemic uncertainty arising from data scarcity, thereby restricting deployment in formally risk-quantified environments. Furthermore, despite random re-sampling of conditioning labels from the training distribution during synthesis, the rarest degradation transitions remain underrepresented in the augmented training set, potentially constraining early-warning performance in the most critical operational regimes. Future research will investigate the integration of Monte Carlo Dropout and Bayesian output layers to provide calibrated confidence intervals for RUL and HI estimates, enabling risk-aware maintenance scheduling in safety-critical contexts. Additional directions include adaptive cGAN retraining strategies to handle concept drift over extended operational lifespans, extension of the monitoring horizon beyond PLC retention limits through transfer learning from related assets, and cross-asset validation across heterogeneous indus-

trial robotic systems to establish broader generalisation boundaries.

8. Statements and Declarations

- **Data and code Availability:** The simulation data used in this study are available through the NASA Prognostics Data Repository at <https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data> Last accessed on 2 June 2026.

The industrial case study data used in this research are subject to strict confidentiality constraints; the participating company has been anonymized at their request, and they have explicitly requested that the raw data not be distributed externally. Consequently, these data are not publicly available. However, to support reproducibility, the full implementation of the GAN-ATCN framework, including the experimental processing pipeline, can be provided by the corresponding author upon reasonable request.

- **Ethics Approval:** Not applicable. This study utilizes simulated data and anonymized technical case study data that do not involve human participants or animals.

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